Research Project: Biomechanical Risk Assessment of Outpatient Physiotherapists

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Methodology: REBA Protocol (Rapid Entire Body Assessment)

ABSTRACT

Physiotherapists working in outpatient settings are exposed to a high risk of Work-related Musculoskeletal Disorders (WMSDs) due to the intrinsic nature of their work: sustained static postures, mobilization of patients with varying degrees of disability, and the application of repetitive manual forces. While the risks in inpatient settings are well-documented, outpatient activity presents specific dynamics—such as prolonged manual treatments, constant adjustment of treatment tables, and the management of assistive devices—that require targeted analysis.

This study aims to quantify the biomechanical risk to which outpatient physiotherapists are subjected. Despite their anatomical expertise, the high frequency of manual mobilizations and the physical demands of the clinical setting necessitate an evidence-based approach to implement preventive measures and optimize ergonomics.

STUDY OBJECTIVES

1. **Quantify** the level of biomechanical risk during common daily clinical activities (manual therapy, therapeutic exercise, patient transfer assistance).
2. **Identify** critical tasks that exceed ergonomic safety thresholds.
3. **Propose** preventive strategies (e.g., setting optimization, use of assistive devices, implementation of active breaks) to reduce the incidence of occupational injuries.

METHODOLOGY: REBA KEY

The study will adopt a descriptive observational approach. Participants will be evaluated during treatment sessions using the REBA method, a validated international tool for whole-body risk analysis. We will monitor the flexion/extension angles of the trunk, neck, and limbs, cross-referencing them with the magnitude of the load handled and the duration of the effort. Data will be collected anonymously and processed in accordance with the GDPR (EU 2016/679).

REBA Parameters:

GROUP A (Trunk/Neck/Legs)	**GROUP B (Arm/Forearm/Wrist)**
TRUNK: 1: Upright; 2: 0-20° F/E; 3: 20-60° F / >20° E	**UPPER ARM:** 1: 20° E-20° F; 2: 20-45° F / >20° E; 3: 45-90° F; 4: >90° F
NECK: 1: 0-20° F; 2: >20° F/E (+1 if twisted)	**FOREARM:** 1: 60-100° F; 2: <60° or >100° F

GROUP A (Trunk/Neck/Legs)	GROUP B (Arm/Forearm/Wrist)
LEGS: 1: Bilateral/Stable; 2: Unstable/Weight shifting	**WRIST:** 1: 0-15° F/E; 2: >15° F/E (+1 if deviated)

Load/Activity Modifiers:

- **LOAD:** +0 (<5 kg), +1 (5-10 kg), +2 (>10 kg).
- **COUPLING:** Scale 0–3 (0 = Good, 3 = Unacceptable).
- **ACTIVITY SCORE:** +1 if static >1 min or repetitive >4 times/min.

INFORMED CONSENT & PRIVACY

Participants agree to be observed/recorded for scientific purposes. Data will be processed anonymously using a unique ID code. Videos will be used exclusively for calculating REBA scores and will be stored on protected servers, accessible only to the research team, in full compliance with the GDPR. Participants retain the right to withdraw from the study at any time.

EXPECTED OUTCOMES

- **Enhanced Safety:** Identification of specific postural criticalities in the outpatient environment.
- **Risk Mitigation:** Evidence-based data to support the reduction of absenteeism due to occupational WMSDs.
- **Ergonomic Optimization:** Guidelines for clinical setting management (e.g., equipment positioning).

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